

VulMorph-Fed: Cross-Project Software Vulnerability Detection via Federated Vulnerability Morphology Learning

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Abstract

Deep learning-based software vulnerability detection faces two coupled barriers: models fail to generalise across projects because they overfit project-specific lexical patterns, and organisations cannot pool the proprietary code that would diversify training data. We propose VulMorph-Fed, a federated learning framework built on the observation that vulnerability manifestations vary lexically across projects while their structural characteristics (morphologies) are largely shared. VulMorph-Fed combines three components: Vulnerability-Critical Sub-graph Abstraction (VCSA), which maps code tokens onto a small, fully specified taxonomy of abstract operation types and learns to down-weight vulnerability-irrelevant edges; Morphology-Conditioned Federated Prototype Aggregation (MCFPA), which shares CWE-conditioned prototypes instead of model parameters; combined with DP-SGD on the local encoders this yields an *end-to-end* (ϵ, δ) -differential-privacy guarantee over the whole pipeline — not merely over the released artefact — accounted with a Rényi accountant; and Morphology-Guided Message Passing (MGMP), which injects the aggregated global prototypes into local graph neural network training. We evaluate on project-level cross-project splits of four public C/C++ corpora (Devign, BigVul, DiverseVul, PrimeVul) against baselines organised by privacy class and matched in data, budget, backbone and input representation, reporting mean and standard deviation over repeated runs with statistical tests. We compare against DP-FedAvg at an identical per-round budget, report both F1 and AUC, and communicate ~ 11 KB per client per round. We also report, candidly, where the design loses: non-private

parameter averaging on the same abstracted graphs remains the stronger aggregator when no privacy constraint is imposed. The complete implementation and scripts regenerating every reported number are publicly released.

Keywords: Program Representation, Static Analysis, Abstract Syntax Trees, Software Vulnerability Detection, Federated Learning, Graph Neural Networks, Differential Privacy

1. Introduction

Software vulnerabilities are a critical and growing threat to modern computing systems. The National Vulnerability Database (NVD) reported over 25,000 new Common Vulnerabilities and Exposures (CVEs) in 2024 alone, highlighting the urgent need for automated and scalable vulnerability detection mechanisms. While Deep Learning-based Vulnerability Detection (DLVD) methods have demonstrated significant promise, they continue to suffer from two fundamental barriers that impede their adoption in real-world scenarios: the generalisation gap and the data silo problem.

First, state-of-the-art DLVD models exhibit a catastrophic *generalisation gap* when applied across different software projects [1]. For instance, a model trained on the source code of FFmpeg typically fails to reliably detect vulnerabilities in OpenSSL, even when both projects manifest the identical Common Weakness Enumeration (CWE) type. Replication studies consistently report performance drops of 50% to 90% in cross-project evaluation settings compared to within-project evaluations [2, 3]. This degradation occurs because existing models tend to overfit to project-specific lexical patterns (e.g., API names, variable naming conventions) rather than learning the underlying structural semantics of the vulnerability.

Second, the *data silo problem* prevents the aggregation of sufficient training data. Real-world vulnerability data is highly sensitive and routinely proprietary. Organisations such as financial institutions, cloud service providers, and defence contractors cannot legally or practically share their raw source code with

external entities due to intellectual property concerns and privacy regulations.

25 Consequently, a centralised DLVD model trained on one organisation’s codebase cannot leverage the knowledge embedded in another organisation’s repositories, severely restricting the model’s exposure to diverse vulnerability manifestations.

Federated Learning (FL) has emerged as a promising paradigm to address the data silo problem by enabling collaborative model training without centralising raw data [4, 5]. However, naively applying standard FL techniques, 30 such as FedAvg, to vulnerability detection does not resolve the generalisation gap [6]. Weight averaging across highly heterogeneous, non-IID (Independent and Identically Distributed) codebases fails to transfer structural vulnerability knowledge effectively, leading to suboptimal cross-project detection performance 35 [7, 8].

The root cause of both barriers stems from the representation level: existing approaches primarily model code at the lexical or token level rather than at the structural or semantic level [9, 10]. A buffer overflow in FFmpeg and a buffer overflow in OpenSSL are structurally identical within their Code Property 40 Graphs (CPGs)—both typically involve a loop incrementing an index, an array access using that index, and a missing bounds check—yet they utilise completely different function names and API calls.

To overcome these challenges, we propose **VulMorph-Fed** (Vulnerability Morphology-Guided Federated Learning), a novel federated learning framework 45 designed to achieve robust cross-project vulnerability detection while preserving data privacy. VulMorph-Fed operates on the principle that vulnerability patterns are structurally invariant across projects but lexically heterogeneous. By extracting and federating *vulnerability morphologies* rather than raw parameters or lexical features, VulMorph-Fed successfully bridges the generalisation 50 gap without requiring centralised code sharing.

The primary contributions of this paper are:

- We introduce the **Vulnerability-Critical Subgraph Abstraction (VCSA)**, which combines a fully specified, deterministic mapping of code tokens

55 onto a small taxonomy of abstract semantic operation types (given exhaustively in Table 1) with a learned, trainable edge-weighting module that down-weights vulnerability-irrelevant structure.

- We propose **Morphology-Conditioned Federated Prototype Aggregation (MCFPA)**, an FL aggregation mechanism that shares CWE-conditioned semantic prototypes instead of model parameters. Combining this with DP-SGD on the local encoders, we prove an *end-to-end* (ϵ, δ) -differential-privacy guarantee (Theorem 1) covering both the transmitted prototypes and the locally trained model, with explicit norm clipping, per-class noise calibration, Rényi accounting of the training budget, and composition over rounds. We are explicit that a prototype guarantee alone would be conditional on a non-private encoder, and we state the residual leakage channels our analysis does not cover.
- We design **Morphology-Guided Message Passing (MGMP)** to inject global prototype signals into local Graph Neural Network (GNN) training, allowing clients to benefit from structural knowledge of CWE types they have never observed locally.
- We evaluate on project-level cross-project splits of four public corpora with multi-seed repetition, statistical testing, and a baseline suite organised by *privacy class*: methods with a formal ϵ -DP guarantee (DP-FedAvg at our exact per-round budget) are compared as peers, while centralised training and parameter-sharing FedAvg — including FedAvg on our own abstracted representation — serve as non-private references. This design isolates the representation from the federation mechanism and prices privacy explicitly. We release the complete implementation and the scripts that regenerate every reported number.¹

80 The remainder of this paper is structured as follows. Section 2 reviews re-

¹<https://github.com/vinhqdang/vulmorph-fed>

lated work in deep learning for vulnerability detection, cross-project prediction, and federated learning. Section 3 details the proposed VulMorph-Fed framework, including its label model, privacy analysis, and inference protocol. Section 4 presents the experimental setup and results. Section 5 discusses implications and limitations, and Section 6 concludes.

2. Background and Related Work

2.1. Deep Learning for Software Vulnerability Detection

The application of Deep Learning (DL) to Software Vulnerability Detection (SVD) has evolved significantly from early sequence-based models to modern graph-based and transformer-based architectures. Early works, such as VulDeePecker [9] and SySeVR [11], treated code as sequences of tokens, employing recurrent neural networks (RNNs) like BiLSTMs to classify code gadgets. While these sequence-based models demonstrated the feasibility of DL for SVD, they struggled to capture the complex, non-linear structural dependencies inherent in source code.

The introduction of the Code Property Graph (CPG) [12], which unifies the Abstract Syntax Tree (AST), Control Flow Graph (CFG), and Data Flow Graph (DFG), marked a paradigm shift. Graph Neural Networks (GNNs) became the architecture of choice for SVD because they naturally align with the topological structure of CPGs. Devign [10] pioneered this approach by applying Gated Graph Neural Networks (GGNNs) over composite graphs to detect vulnerabilities in large C/C++ projects. Subsequent GNN-based models have focused on improving graph representation and interpretability. For instance, DeepWukong [13] operates on inter-procedural program slices, while IVDetect [14] provides fine-grained interpretations using GNNExplainer. More recent advancements include Vul-LMGNNs [15], which fuses Large Language Models (LLMs) with GNNs, and HAGNN [16], which utilises heterogeneous attention mechanisms.

Simultaneously, Transformer-based models and LLMs have been adapted for SVD. LineVul [17] leverages CodeBERT for line-level vulnerability prediction,

110 and LLMxCPG [18] uses CPGs to guide LLM attention. Despite these advances, recent studies [19, 20] indicate that LLMs often operate at a shallow structural level, highlighting the continued necessity for deep structural abstractions. Furthermore, rigorous replication studies [1, 2, 3] have consistently exposed a severe generalisation gap: SOTA models perform well in within-project settings but
115 their accuracy drops precipitously when evaluated on unseen projects (cross-project).

2.2. Cross-Project Vulnerability and Defect Prediction

Cross-Project Vulnerability Detection (CPVD) and Cross-Project Defect Prediction (CPDP) aim to train models on a source project and deploy them on
120 a target project. Traditional CPDP often relies on software metrics and transfer learning [7]. In the context of DLVD, several works have employed Domain Adaptation (DA) techniques. CD-VulD [21] and VulGDA [22] use deep domain adaptation on graph embeddings to achieve cross-domain discovery. CPVD [23] combines Graph Attention Networks (GATs) with adversarial domain adap-
125 tation, representing the current SOTA for fully supervised cross-project SVD. CSVD-TF [24] fuses expert and semantic metrics using TrAdaBoost. However, a critical limitation of these DA-based approaches is that they require centralised access to raw source code from both the source and target domains, violating privacy constraints in many real-world scenarios.

130 2.3. Federated Learning in Software Engineering

Federated Learning (FL) [4] allows multiple clients to jointly train a global model without sharing their local data. Its application to Software Engineering (FL-SE) is a burgeoning area of research, motivated by the proprietary nature of commercial codebases [25]. Recent empirical studies [26, 8] confirm the fea-
135 sibility of FL for tasks like code clone detection and vulnerability detection.

However, adapting FL to SVD presents unique challenges, primarily due to severe non-IID data distributions (heterogeneity in CWE types and coding styles). FedGraphNN [6] demonstrated that standard federated GNNs often

underperform centralised models on non-IID graph splits. Existing FL-SVD
 140 solutions typically rely on NLP models (e.g., CodeBERT) or naive GNN aggregations [27, 8], which inherit the generalisation gap of their centralized counterparts. To address graph heterogeneity, approaches like FedDense [28] decouple structural and feature learning. VulMorph-Fed builds upon these concepts but specifically targets the structural invariance of vulnerabilities by federating semantic
 145 prototypes rather than raw model parameters.

2.4. Privacy-Preserving Mechanisms

Ensuring data privacy in FL often involves Differential Privacy (DP) [5] or Secure Aggregation [29]. DP algorithms, such as DP-SGD, add calibrated noise (e.g., Laplace or Gaussian) to model updates. The magnitude of this noise is proportional to the sensitivity of the function being computed. Standard DP-SGD
 150 on large GNNs or LLMs requires significant noise due to the high dimensionality of the model parameters and the large vocabulary size of source code tokens, severely degrading utility. VulMorph-Fed mitigates this by applying DP at the level of semantic prototypes. By utilizing a compact, abstracted morphological space instead of the full token vocabulary, the sensitivity is drastically reduced,
 155 enabling a much more favorable privacy-utility trade-off.

3. Methodology

To address the cross-project generalisation gap and the data-silo problem simultaneously, we propose *VulMorph-Fed*. The framework replaces standard parameter averaging with a privacy-preserving federated aggregation of semantic
 160 structural prototypes. It consists of three components: (1) Vulnerability-Critical Subgraph Abstraction (VCSA), which maps functions onto project-invariant structural morphologies; (2) Morphology-Conditioned Federated Prototype Aggregation (MCFPA), which aggregates structural knowledge under differential privacy; and (3) Morphology-Guided Message Passing (MGMP), which injects
 165 the aggregated global knowledge into local graph learning. Figure 1 illustrates

the end-to-end abstraction on two lexically different but structurally similar vulnerable functions.

3.1. Graph Construction

170 Let $\mathcal{G}_k = \{(G_i^{(k)}, y_i^{(k)}, c_i^{(k)})\}_{i=1}^{N_k}$ denote the local dataset of client $k \in \{1, \dots, K\}$, where $y_i \in \{0, 1\}$ is the binary vulnerability label and c_i the CWE annotation of vulnerable functions (Section 3.3).

Full Code Property Graph (CPG) extraction requires a heavyweight static-analysis engine such as Joern [12], which is impractical to deploy at every fed-
 175 erated edge client and complicates exact reproduction. Our implementation instead parses every function with `tree-sitter` — a production-grade, error-tolerant C/C++ parser — and builds the graph $G = (\mathcal{V}, \mathcal{E})$ directly from the abstract syntax tree:

- one node per *named AST node*, in pre-order, capped at $|\mathcal{V}| \leq 200$;
- 180 • *syntax edges* $\mathcal{E}_{AST}$: parent-child links in both directions;
- *order edges* $\mathcal{E}_{SIB}$: consecutive named siblings (source-order sequencing, the AST analogue of Devign’s NCS edges [10]);
- *data-dependence proxy edges* $\mathcal{E}_{DD}$: def-use chains linking successive occurrences of the same identifier leaf.

185 The edge set is $\mathcal{E} = \mathcal{E}_{AST} \cup \mathcal{E}_{SIB} \cup \mathcal{E}_{DD}$; on our corpora this yields on average ~ 130 nodes and ~ 350 edges per function. Functions that fail to parse (a small minority) fall back to a token-level sequence graph and are marked as such. These graphs carry real syntactic structure, though not the interprocedural control- and data-flow precision of a compiler-grade CPG; Section 5
 190 discusses this trade-off, and graph construction is isolated behind a single interface so Joern-derived CPGs can be substituted without touching the rest of the framework.

3.2. Vulnerability-Critical Subgraph Abstraction (VCSA)

Raw code graphs carry thousands of project-specific API and identifier tokens, causing GNNs to overfit local lexical distributions. VCSA replaces the
195 lexical node labels with a small taxonomy of abstract operation types and learns to down-weight vulnerability-irrelevant edges.

3.2.1. Morphological taxonomy

We define a fixed semantic taxonomy $\mathcal{T}$ of $|\mathcal{T}| = 8$ operation types, listed
200 exhaustively in Table 1 together with the reserved fallback type UNKNOWN. A node therefore takes one of $|\mathcal{T}| + 1 = 9$ morphological categories — the UNKNOWN type has its own feature slot. For the taxonomy-size sensitivity analysis (RQ2b) we additionally define strictly nested refinements with $|\mathcal{T}| = 16$ and $|\mathcal{T}| = 32$ (e.g., MEMORY_ACCESS splits into MEMORY_ALLOC, MEMORY_FREE, MEMORY_COPY,
205 STRING_OP); the complete lists ship with the source code.

Each node’s feature vector is the sum of two embeddings, both drawn from *project-invariant* vocabularies: (i) its morphological type in $\mathcal{T} \cup \{\text{UNKNOWN}\}$, and (ii) its *grammar kind* — the tree-sitter node category (`call_expression`, `subscript_expression`, `if_statement`, ...), a fixed set of 364 categories de-
210 termined entirely by the C grammar and therefore identical for every project and client by construction. No identifier, API name, literal or any other project-specific token ever enters the network.

3.2.2. The abstraction function ϕ_{type}

The mapping $\phi_{\text{type}} : \mathcal{V} \rightarrow \mathcal{T} \cup \{\text{UNKNOWN}\}$ is a deterministic rule set evaluated
215 on the AST, in priority order:

1. a `call_expression` is classified by its callee name in two stages: an exact match against fixed libc allow-lists (`malloc`, `memcpy`, `strcpy`, ...), then a set of *affix rules* that recognise project-wrapped equivalents. The affix stage is essential rather than cosmetic: production C code almost never
220 calls libc allocators directly (FFmpeg uses `av_malloc`, GLib `g_malloc`, the Linux kernel `kmalloc`, OpenSSL `OPENSSL_malloc`), so an exact-match list

Table 1: The complete $|\mathcal{T}| = 8$ morphological taxonomy (plus the reserved UNKNOWN fallback). A node takes one of $|\mathcal{T}| + 1 = 9$ morphological *categories*; each category is mapped to a learned embedding (dimension 64 in our configuration), summed with the node’s grammar-kind embedding.

Abstract type	Covers
MEMORY_ACCESS	allocation/free/copy/set and string APIs (<code>malloc</code> , <code>free</code> , <code>memcpy</code> , <code>strcpy</code> , ...)
ARRAY_INDEX	array subscript operations <code>a[i]</code>
PTR_DEREF	pointer dereference and member access (<code>*p</code> , <code>p->f</code> , unary <code>&</code>)
CONTROL_BRANCH	<code>if/else/switch</code> , loops, <code>goto/break/continue/return</code> , ternary
ARITH_OP	arithmetic and bitwise operators (<code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>%</code> <code>&</code> <code> </code> <code>^</code> <code>~</code> <code><<</code> <code>>></code>)
COMPARISON	relational and logical operators (<code>==</code> <code>!=</code> <code><</code> <code>></code> <code><=</code> <code>>=</code> <code>&&</code> <code> </code> <code>!</code>)
CALL_SITE	any other function/macro call (user-defined and unresolved calls included)
ASSIGN	plain and compound assignments (<code>=</code> , <code>+=</code> , ...)
UNKNOWN	all remaining tokens (identifiers, literals, punctuation, declarations)

alone routes nearly every real allocation site to `CALL_SITE`. We measured this: on Devign, exact matching alone assigns `MEMORY_ACCESS` to only 0.23% of AST nodes, rising to 0.72% once affix rules are applied — i.e. two thirds of the memory-safety signal the taxonomy is designed to capture is invisible without them. The rules test for the affixes `*alloc*`, `*free*`, `*memcpy*`, `*memset*`, `*str{cpy,cat,len}*` and are evaluated in an order that resolves `realloc` and `dealloc` before the generic `alloc` test. Only the matched *class* is retained — never the identifier — so project invariance is preserved. Unmatched calls, including all user-defined functions, macros and unresolved calls, map to `CALL_SITE`; no name resolution or build system is required;

2. statement and expression kinds map directly: `subscript_expression` $\rightarrow$ `ARRAY_INDEX`; `field_expression` and unary pointer expressions $\rightarrow$ `PTR_DEREF`; `if/switch/loop/jump` statements and ternaries $\rightarrow$ `CONTROL_BRANCH`; `assignment_expression` and initialising declarators $\rightarrow$ `ASSIGN`;
3. a `binary_expression` maps by its operator text to `ARITH_OP` or `COMPARISON` (Table 1);
4. three further kind rules complete the set: `cast_expression` $\rightarrow$ `PTR_DEREF`

240 (a cast reinterprets a pointee type), `update_expression` (i++, i-) $\rightarrow$
 ARITH_OP, and unary ! $\rightarrow$ COMPARISON with the remaining unary opera-
 tors $\rightarrow$ ARITH_OP;
 5. every remaining node maps to UNKNOWN.

An equivalent token-context rule set (documented in the code) covers the rare
 245 parse-failure fallback graphs. The rules are implemented once at the finest
 ($|\mathcal{T}| = 32$) granularity and projected onto the coarser taxonomies, so all three
 taxonomy sizes share a single rule set. ϕ_{type} is a discrete label assignment and is
not differentiable; the only learned part of VCSA is the edge-weighting module
 below.

250 On our four evaluation corpora, 25–26% of AST nodes receive a non-UNKNOWN
 morphological type (the remainder are declarations, identifiers, literals and other
 syntax carriers, which still contribute their grammar-kind feature). The abstrac-
 tion does not delete nodes: all nodes are retained with their graph structure, but
 their features are collapsed from a $\sim 10,000$ -token lexical vocabulary onto $|\mathcal{T}| + 1$
 255 morphological categories plus 364 grammar kinds. This many-to-one collapse
 is what removes project-specific lexical information from everything the model
 computes and, downstream, from everything a client transmits (Section 3.4.1).

3.2.3. Learned edge weighting

VCSA additionally learns a soft edge-importance mask. A two-layer MLP
 260 over concatenated endpoint features produces

$$\varepsilon_e = \sigma(\text{MLP}([\mathbf{x}_u \parallel \mathbf{x}_v])) \in [0, 1] \quad \text{for } e = (u, v) \in \mathcal{E}, \quad (1)$$

which reweights messages during propagation (Eq. 7); edges are softly down-
 weighted rather than hard-pruned, keeping the pipeline trainable end-to-end.
 Sparsity is encouraged by an ℓ_1 penalty $\mathcal{L}_{\text{sparse}} = \frac{1}{|\mathcal{E}|} \sum_e \varepsilon_e$. To align mor-
 phologies across clients we further use a structural contrastive loss over graph

Project A (FFmpeg-style):

```
n = av_get_size(ctx);
buf = av_malloc(n);
for (i = 0; i <= n; i++)
    buf[i] = src[i];
```

Project B (OpenSSL-style):

```
len = BN_num_bytes(a);
out = OPENSSL_malloc(len);
for (k = 0; k <= len; k++)
    out[k] = in[k];
```

Morphology sequence emitted by ϕ_{type} — identical for both fragments:

```
CONTROL_BRANCH → ASSIGN → ASSIGN → ASSIGN → COMPARISON → ARITH_OP → CALL_SITE →
MEMORY_ACCESS → ASSIGN → ARRAY_INDEX → ARRAY_INDEX
```

Figure 1: Worked example of the VCSA abstraction. Two off-by-one buffer overflows from different projects share no API or identifier token — `av_get_size`/`BN_num_bytes`, `av_malloc`/`OPENSSL_malloc`, `buf`/`out` — yet ϕ_{type} maps both onto a byte-identical morphology: a size query (`CALL_SITE`), an allocation (`MEMORY_ACCESS`), and a loop whose *inclusive* bound (`COMPARISON` on $\leq$) drives an array write (`ARRAY_INDEX`). The sequence shown is the literal output of the released `classify_token/_classify_ast_node` implementation on these two inputs (breadth-first over the AST, `UNKNOWN` nodes elided), not an idealisation; it is regenerated from the code so that figure and implementation cannot diverge. Note that recognising `av_malloc` and `OPENSSL_malloc` as allocations requires the affix rules of Section 3.2 — an exact libc allow-list maps both to `CALL_SITE` and the shared morphology is lost.

265 embeddings $\mathbf{h}_G$:

$$\mathcal{L}_{SCL} = \frac{1}{|\mathcal{P}^+ \cup \mathcal{P}^-|} \left[\sum_{(i,j) \in \mathcal{P}^+} \|\mathbf{h}_{G_i} - \mathbf{h}_{G_j}\|_2^2 + \sum_{(i,j) \in \mathcal{P}^-} \max(0, m - \|\mathbf{h}_{G_i} - \mathbf{h}_{G_j}\|_2^2) \right], \quad (2)$$

where $\mathcal{P}^+$ are same-CWE vulnerable pairs and $\mathcal{P}^-$ are (vulnerable, benign) pairs within a mini-batch, with margin $m = 1$. The total local objective is $\mathcal{L} = \mathcal{L}_{BCE} + \alpha \mathcal{L}_{SCL} + \gamma \mathcal{L}_{\text{sparse}}$ with $\alpha = 0.1$, $\gamma = 0.01$; $\mathcal{L}_{BCE}$ is class-weighted binary cross-entropy with `pos_weight` = $\min(N^-/N^+, 20)$ computed on each
270 client’s local split.

3.3. Label Model

Real vulnerability corpora have a many-to-many CVE–function–CWE structure, so we state our label model explicitly:

- **Detection is strictly binary.** The classification head predicts $\hat{y} \in [0, 1]$ (vulnerable vs. benign); all reported Precision, Recall, F1 and AUC values

275

are computed from this binary head. The decision threshold is calibrated on held-out training-project data as described in Section 4, never on the test set; AUC is threshold-free. CWE information never enters the detection head.

- 280 • **CWE labels condition only the prototypes.** A vulnerable function annotated with several CWEs contributes to the prototype of its *first-listed* (*primary*) CWE. Benign functions contribute to a dedicated *benign* prototype slot, giving the bank an explicit model of non-vulnerable morphology.
- 285 • **CWE vocabulary.** The prototype bank has $|\mathcal{C}| + 1 = 11$ slots: the nine most frequent CWE types in the training corpus receive dedicated slots, all remaining types share an **OTHER** slot, and the eleventh slot is the benign prototype. This bucketing is computed once per dataset and reported with the released configurations. Datasets without per-function
- 290 CWE annotations (Devign) degenerate gracefully to a single vulnerable-prototype slot plus the benign slot.

3.4. Morphology-Conditioned Federated Prototype Aggregation (MCFPA)

FedAvg-style parameter averaging [4] assumes near-IID clients; vulnerability data is extremely non-IID in its CWE composition. MCFPA instead federates

295 *class-conditioned prototypes* over the $|\mathcal{C}| + 1$ bank slots of Section 3.3 (CWE buckets plus the benign slot). After local training, client k computes, for each slot c with local support $N_{c,k} > 0$,

$$\mathbf{p}_c^{(k)} = \frac{1}{N_{c,k}} \sum_{G \in \mathcal{G}_k: y_G=1, c_G=c} \text{clip}_R\left(f_{\theta_k}(\tilde{G})\right), \quad \text{clip}_R(\mathbf{z}) = \mathbf{z} \cdot \min\left(1, \frac{R}{\|\mathbf{z}\|_1}\right), \quad (3)$$

i.e., the centroid of the ℓ_1 -clipped embeddings of its local functions assigned to slot c (vulnerable functions of the corresponding CWE bucket; all benign

300 functions for the benign slot). Slots with $N_{c,k} = 0$ are transmitted as zero rows and carry no information.

3.4.1. Differential privacy

Before upload, each non-empty prototype row is perturbed with the Laplace mechanism:

$$\hat{\mathbf{p}}_c^{(k)} = \mathbf{p}_c^{(k)} + \text{Lap}\left(0, \frac{2R}{N_{c,k} \varepsilon}\right)^{\otimes d}. \quad (4)$$

305 VulMorph-Fed applies two mechanisms, so that the guarantee covers the whole pipeline rather than only the released artefact.

Mechanism 1: private encoder training.. Local training uses DP-SGD [5]: each per-example gradient is clipped to $\|g_i\|_2 \leq C$ and the batch sum is perturbed with Gaussian noise of standard deviation σC before the optimiser step. The
310 encoder θ_k is therefore itself a differentially private function of the client’s data. This matters: without it, everything below would be conditional on an encoder that had seen the raw data unprotected, and changing one record could move every embedding by an unbounded amount.

Mechanism 2: private prototype release.. Prototypes are built from ℓ_1 -clipped
315 embeddings (Eq. 3) and perturbed per class (Eq. 4).

Proposition 1 (Per-round privacy of the prototype release). *Fix a client k and condition on its (DP-trained) encoder. Let two neighbouring datasets differ in one function. Releasing the noisy bank $\{\hat{\mathbf{p}}_c^{(k)}\}$ of Eq. (4) satisfies 2ε -differential privacy with respect to that record.*

320 *Proof.* With the encoder fixed, each per-sample embedding lies in the ℓ_1 ball of radius R , so replacing one record changes an affected class mean by at most $\Delta_1 = 2R/N_{c,k}$ in ℓ_1 norm; the Laplace mechanism with scale Δ_1/ε is ε -DP for that row [30]. Under the replacement neighbouring relation a record change can affect *two* rows — it leaves the slot of its old label and joins the slot of its new
325 one — so parallel composition across slots does not apply and the two affected rows compose sequentially, giving 2ε . (A restricted neighbouring relation that preserves the label would give ε ; we report the weaker, safer constant.) $\square$

Theorem 1 (End-to-end budget). *Let ε_{sgd} be the budget consumed by DP-SGD over all local steps, accounted under Rényi differential privacy for the subsampled Gaussian mechanism [31] at a target δ . Then T rounds of VulMorph-Fed satisfy*

$$\varepsilon_{\text{tot}} = \varepsilon_{\text{sgd}} + 2T\varepsilon \quad (5)$$

($\varepsilon_{\text{tot}}, \delta$)-differential privacy with respect to any single function in a client’s dataset, covering both the transmitted prototypes and the locally trained encoder.

Proof. DP-SGD over the local steps is $(\varepsilon_{\text{sgd}}, \delta)$ -DP by the Rényi accountant; the T prototype releases are $2T\varepsilon$ -DP by Proposition 1 and sequential composition. The two mechanisms observe the same private dataset, so their budgets add. $\square$

We use the standard Rényi accountant (the binomial bound of Mironov et al. [31] for the sampled Gaussian, converted to (ε, δ) over a grid of orders) rather than advanced composition, which would overstate ε_{sgd} by more than an order of magnitude; our implementation reproduces the reference regime of Abadi et al. [5] to within the expected margin.

Residual leakage outside the analysis.. Two channels remain, and we state them rather than leaving them implicit. The noise scale $2R/(N_{c,k}\varepsilon)$ depends on the private count $N_{c,k}$, and the *support pattern* of a client’s occupied slots is transmitted unnoised, revealing which weakness classes a client possesses at all. Padding the bank to a fixed support and using a count-independent scale would close both at some utility cost; we quantify neither here.

We emphasise why the morphological abstraction matters here: because the encoder consumes only $|\mathcal{T}|+1$ abstract node types, the embeddings being clipped and averaged contain no project-specific lexical information to begin with. DP protects *membership* of individual functions; the abstraction removes *content* (identifiers, API names) categorically rather than statistically.

3.4.2. Affinity-weighted aggregation

The server aggregates the noisy banks per CWE bucket. Let $A_k \subseteq \{1..K\}$ be the clients with non-zero rows for bucket c . With $\bar{\mathbf{p}}_j = \hat{\mathbf{p}}_c^{(j)} / \|\hat{\mathbf{p}}_c^{(j)}\|_2$, the

355 server computes cosine-affinity weights and the global prototype

$$w_c^{(k)} = \max\left(0, \frac{1}{|A_k| - 1} \sum_{j \in A_k, j \neq k} \bar{\mathbf{p}}_j^\top \bar{\mathbf{p}}_k\right), \quad \mathbf{p}_c^{\text{glob}} = \frac{\sum_{k \in A_k} w_c^{(k)} \hat{\mathbf{p}}_c^{(k)}}{\sum_{k \in A_k} w_c^{(k)}}, \quad (6)$$

falling back to the uniform mean when all affinities vanish, and to the previous round’s value when no client has data for c . Clients whose local view of CWE c agrees with the consensus contribute more; outlier (or heavily noised) views are attenuated. The ablation “w/o MCFPA” replaces Eq. (6) with the uniform
360 mean.

3.5. Morphology-Guided Message Passing (MGMP)

Each client integrates the broadcast bank $\mathcal{P}^{\text{glob}} = \{\mathbf{p}_c^{\text{glob}}\}$ into its GNN. Layer l first performs edge-weighted local aggregation (GCN-style normalisation $\hat{d}$ with the VCSA mask ε as edge weight):

$$\mathbf{h}_v^{\text{loc}} = \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{\varepsilon_{uv}}{\sqrt{\hat{d}_u \hat{d}_v}} \mathbf{W}^{(l)} \mathbf{h}_u^{(l)}, \quad (7)$$

365 then attends over the global prototypes with scaled dot-product attention:

$$\mathbf{m}_v = \sum_{c \in \mathcal{C}} \text{softmax}_c \left(\frac{(\mathbf{W}^{(l)} \mathbf{h}_v^{(l)})^\top \mathbf{p}_c^{\text{glob}}}{\sqrt{d}} \right) \mathbf{p}_c^{\text{glob}}, \quad (8)$$

and fuses the two signals through a per-node morphology gate $\beta_v = \sigma(\mathbf{w}_\beta^\top [\mathbf{W}^{(l)} \mathbf{h}_v^{(l)} \parallel \mathbf{x}_v^{\text{morph}}])$ and a learnable per-layer balance $\lambda \in [0, 1]$:

$$\mathbf{h}_v^{(l+1)} = \text{ReLU}((1 - \lambda) \mathbf{h}_v^{\text{loc}} + \lambda \beta_v \mathbf{m}_v). \quad (9)$$

Nodes whose abstract type is implicated in known vulnerability morphologies receive a higher gate β_v and thus a stronger global signal. Even a client that
370 has never observed CWE c locally can recognise its structural morphology by attending to $\mathbf{p}_c^{\text{glob}}$.

The global bank additionally reaches the decision through two further channels. First, the graph-level readout is $[\mathbf{h}_G^{\text{mean}} \parallel \mathbf{h}_G^{\text{max}} \parallel \cos(\mathbf{h}_G^{\text{mean}}, \mathbf{p}_c^{\text{glob}})_{c \in \mathcal{C} \cup \{\text{ben}\}}]$:

the classifier directly sees how similar the current function is to every global mor-
 375 phology, including the benign one. Second, a prototype-alignment regulariser

$$\mathcal{L}_{\text{proto}} = \frac{1}{|B|} \sum_{i \in B} \|\mathbf{h}_{G_i} - \mathbf{p}_{s(i)}^{\text{glob}}\|_2^2, \quad (10)$$

where $s(i)$ is sample i 's bank slot (its CWE bucket, or the benign slot), pulls each
 client's embedding space toward the shared global geometry — the mechanism
 FedProto [32] showed to be essential for prototype-based FL under non-IID
 clients. The total local objective becomes $\mathcal{L} = \mathcal{L}_{BCE} + \alpha \mathcal{L}_{SCL} + \gamma \mathcal{L}_{\text{sparse}} +$
 380 $\mu \mathcal{L}_{\text{proto}}$ with $\mu = 0.5$.

3.6. Training and Inference Protocol

Algorithm 1 summarises one federation. Note that *model parameters are*
never aggregated: each client keeps its own encoder θ_k , and the only shared
 state is the prototype bank.

385 **Inference.** The deployed global detector is the uniform probability ensem-
 ble of the K client models, each conditioned on the same final prototype bank:

$$\hat{p}(y \mid G) = \frac{1}{K} \sum_{k=1}^K \sigma\left(f_{\theta_k}(\tilde{G}; \mathcal{P}^{\text{glob}})\right). \quad (11)$$

All headline metrics are computed from Eq. (11) on the held-out cross-project
 test set; we additionally report the mean and standard deviation of per-client
 F1 to expose client-level variance. In a production deployment, Eq. (11) cor-
 390 responds to clients exposing scoring endpoints, or — when a single model is
 required — any client model can be used alone, at the cost quantified by the
 reported per-client variance.

3.7. Communication Cost

Per round, each client uploads and downloads one $(|\mathcal{C}| + 1) \times d$ float32 bank:
 395 $2 \times 11 \times 128 \times 4 \text{ B} \approx 11 \text{ KB}$ per client per round in our configuration, independent
 of model size; total server traffic is K times that (e.g., $\approx 220 \text{ KB}$ per round at
 $K = 20$). Standard FedAvg on the same backbone transmits the full parameter
 vector ($\sim 1.4 \text{ MB}$ for our deliberately compact GNN; hundreds of MB for code-
 LM backbones) per client per round.

Algorithm 1 VulMorph-Fed

Require: K clients, rounds T , per-round budget ε , clip radius R

```
1: Server initialises  $\mathcal{P}^{\text{glob},(0)} = \mathbf{0}$ 
2: for  $t = 1, \dots, T$  do
3:   for each client  $k$  in parallel do
4:     Retrieve  $\mathcal{P}^{\text{glob},(t-1)}$ ; abstract graphs via VCSA ( $\phi_{\text{type}}$ , Eq. 1)
5:     Train  $\theta_k$  for  $E$  local epochs with MGMP (Eqs. 7–9) and loss  $\mathcal{L}_{BCE} +$   

 $\alpha\mathcal{L}_{SCL} + \gamma\mathcal{L}_{\text{sparse}} + \mu\mathcal{L}_{\text{proto}}$  (Eq. 10)
6:     Build clipped prototypes  $\mathbf{p}_c^{(k)}$  (Eq. 3); perturb via Eq. (4)
7:     Upload  $\{\hat{\mathbf{p}}_c^{(k)}\}_{c \in \mathcal{C}}$   $\{2|\mathcal{C}|d$  floats per round incl. download $\}$ 
8:   end for
9:   Server aggregates  $\mathcal{P}^{\text{glob},(t)}$  via Eq. (6) and broadcasts
10: end for
Ensure: Prototype bank  $\mathcal{P}^{\text{glob},(T)}$ ; ensemble detector of Eq. (11); spent budget  

 $\varepsilon_{\text{tot}} = T\varepsilon$ 
```

400 4. Experimental Evaluation

We design the evaluation to answer five research questions:

- **RQ1 (Cross-Project Generalisation):** How does VulMorph-Fed compare to centralised and federated baselines — including baselines that share its backbone or its input representation — on held-out projects
405 never seen by any client?
- **RQ1b (Value of Federation):** What do private federated clients add on top of a model trained centrally on public data alone?
- **RQ2 (Ablations):** What do VCSA, MCFPA and MGMP contribute individually, and how sensitive is the framework to the taxonomy size
410 $|\mathcal{T}|$?
- **RQ3 (Privacy-Utility):** How does detection quality degrade across per-round DP budgets ε , accounting for composition over rounds?

- **RQ4 (Scalability):** How do performance and communication cost behave as the number of clients K grows?

415 4.1. Experimental Setup

4.1.1. Datasets

We evaluate on four public C/C++ vulnerability corpora with complementary characteristics:

1. **Devign** [10]: 21,854 manually labelled functions from FFmpeg and QEMU (≈46% vulnerable); no per-function CWE annotations.
2. **BigVul** [33]: CVE-linked functions from 234 projects (in our sample), with CWE annotations; realistic class imbalance (≈6% vulnerable).
3. **DiverseVul** [34]: the most project-diverse corpus (580 projects in our sample, 59 CWE types; ≈5.5% vulnerable).
- 425 4. **PrimeVul** [3]: the strictest recent benchmark, with rigorous de-duplication and label verification.

Devign is loaded from its public HuggingFace mirror, BigVul and DiverseVul from the `bstee615` mirrors, and PrimeVul from the `ASSERT-KTH` mirror (unpaired training split); exact dataset revisions are pinned in the released code.

430 For computational tractability every experiment uses the first 8,000 functions of each corpus (after discarding functions shorter than three tokens); we report this cap explicitly since sampling affects absolute numbers.

4.1.2. Cross-project federated split

All splits are at *project* granularity: every function of a repository is assigned to exactly one partition. Projects are sorted by sample count and the smallest projects are held out until they total ≈20% of samples — these held-out projects form the test set, and *no function from any test project is ever seen by any client*. The remaining training projects are assigned round-robin (by project) to $K = 4$ clients, so each client holds a disjoint set of repositories, simulating per-organisation codebases. For BigVul, DiverseVul and PrimeVul this yields

440

hundreds of held-out test projects; for Devign, which contains only two projects, the split degenerates to training on one project and testing on the other (a pure cross-project transfer task). Every RQ1 experiment is repeated with five random seeds $\{42, \dots, 46\}$ (three for the secondary analyses). The seed draws a *fresh*
 445 *project-level holdout* as well as the client assignment, model initialisation and DP noise, so the reported standard deviations include split variance — which, in cross-project evaluation, dominates initialisation variance. We report mean $\pm$ standard deviation throughout, and per-seed values feed the statistical tests below.

450 4.1.3. Baselines

The baseline suite is organised by *privacy class*, and within each class it isolates the two factors that could otherwise confound the comparison — input representation and aggregation mechanism.

Non-private references (raw code pooled, or parameters exchanged in the clear — vulnerable to gradient-inversion and membership attacks, hence
 455 *not* peers of a DP system):

- **Centralised GGNN / GAT (lexical graphs)**: Devign-style [10] and CPVD-style [23] backbones on pooled raw data;
- **Centralised Transformer (sequence)**: a Transformer encoder over
 460 raw token sequences (structure-free family, LineVul-style [17]); a compact model trained from scratch, not a pre-trained code LM — pre-trained LM baselines are left to future work;
- **Centralised GGNN (VCSA graphs)**: the GGNN backbone on our abstracted representation, isolating representation from protocol;
- **FedAvg + GAT / Transformer (lexical)**: standard federated base-
 465 lines;
- **FedAvg + GGNN (VCSA graphs)**: FedAvg on the *same* input representation as VulMorph-Fed;

- **Centralised VulMorph (oracle)**: our full model on pooled data, no federation, no DP.

Formally private peers (per-round ε -DP, matched to VulMorph-Fed’s budget):

- **DP-FedAvg + GAT / + GGNN (VCSA graphs)**: the standard route to formal privacy for parameter-sharing FL — each client’s model update is norm-clipped and perturbed with a Laplace mechanism calibrated to the same per-round ε as our prototype release. Both mechanisms spend the same per-round ε , but they spend it very differently: a single ℓ_1 clipping budget must cover the entire parameter update (0.19–0.67M coordinates depending on backbone), whereas ours covers an $(|\mathcal{C}| + 1) \times d = 1,408$ -coordinate bank whose rows are additionally means over $N_{c,k}$ records. The per-coordinate signal-to-noise ratio therefore differs by orders of magnitude. We report this as the operative mechanism rather than attributing the effect to dimensionality *per se*: for the Laplace mechanism the per-coordinate noise scale is Δ_1/ε and does not depend on d directly.

All baselines use the same data, splits, seeds, class-weighted loss, downsampling, threshold calibration, optimiser and training budget as VulMorph-Fed; FedAvg aggregation is sample-size weighted.

4.1.4. Implementation details

All models are implemented in PyTorch / PyTorch Geometric; graphs are built with tree-sitter as described in Section 3.1. VulMorph-Fed uses a 2-layer GNN, hidden dimension $d = 128$, embedding dimension 64, mean $\oplus$ max readout, dropout 0.3, Adam with learning rate 10^{-3} , batch size 64, $T = 10$ federated rounds with $E = 2$ local epochs, $|\mathcal{C}| = 10$ CWE buckets, taxonomy size $|\mathcal{T}| = 8$, per-round budget $\varepsilon = 2.0$ and clip radius $R = 1$ unless stated otherwise. Two class-imbalance measures apply identically to every system (ours and all baselines): each client’s *training* split caps the benign:vulnerable ratio at 4:1 (test and calibration splits keep true prevalence), and the loss is class-weighted

with $\text{pos_weight} = \min(N^-/N^+, 20)$. Inference follows the ensemble protocol of Eq. (11). The decision threshold of every system is calibrated by maximising
 500 F1 on a calibration slice (10% of each client’s training-project samples, held out before downsampling); *the test set never influences the threshold*, and AUC is threshold-free.

4.1.5. Statistical testing

For each baseline we pair the per-dataset, per-seed F1 values of VulMorph-
 505 Fed with those of the baseline (up to $4 \times 3 = 12$ pairs) and run a two-sided Wilcoxon signed-rank test with Cliff’s δ effect size; we mark a comparison significant at $p < 0.05$. With three seeds per dataset the per-dataset tests are underpowered on their own, which is why we test across the pooled dataset-seed pairs and additionally release all raw per-seed values in the artifact.

510 4.1.6. Reproducibility

The complete implementation — loaders, models, federated runners, base-
 lines, and the scripts that regenerate every table and figure in this paper from scratch (`experiments/run_all.sh`) — is publicly available.² Every table in this
 515 paper is generated programmatically from the versioned result files; no number is entered by hand.

4.2. RQ1: Cross-Project Generalisation

Table 2 and Figure 2 report cross-project F1 on the four corpora, with methods grouped by privacy class. Three observations stand out.

First, *within the formally private class* — the comparison that matches guar-
 520 antee for guarantee — VulMorph-Fed outperforms DP-FedAvg on both representations and on every dataset. The mechanism is visible in the numbers: at the same per-round ϵ , DP-FedAvg must spread its noise over a $\sim 350\text{K}$ -dimensional parameter update and collapses toward chance, while our $\sim 1.4\text{K}$ -dimensional

²<https://github.com/vinhqdang/vulmorph-fed>

prototype release retains most of the non-private performance. The claimed
525 sensitivity advantage of prototype-level DP is thus measured, not asserted.

Second, the *representation* carries transfer on its own: FedAvg + GGNN
on VCSA-abstracted graphs is the strongest federated configuration and sur-
passes even the centralised lexical GGNN’s discrimination on the multi-project
corpora. Lexical features do not transfer across project boundaries; removing
530 them costs little and removes the main source of client drift. The remaining gap
between VulMorph-Fed and *non-private* FedAvg on the same representation is
the measured price of never transmitting parameters — and it is small relative
to the price DP-FedAvg pays for the same formal guarantee.

Third, absolute cross-project F1 remains far from the within-project lev-
535 els reported in the older literature — consistent with the replication stud-
ies [1, 2, 3]. We deliberately do not claim to have “closed” the generalisation
gap; we claim, and the data supports, that under matched privacy budgets
our design is the strongest option, and that its distance to non-private feder-
ation is modest. The pooled Wilcoxon tests against each baseline are released
540 in `results/significance.json`; comparisons that do not reach $p < 0.05$ are
stated as such rather than described as significant.

4.3. RQ1b: What Does Federation Add Beyond Public Data?

A fair question is whether federation is needed at all when large public
vulnerability corpora exist: if morphologies are project-invariant, a centralised
545 model on public data might transfer to private codebases without any federation.
Table ?? addresses this directly. We train (A) the full VulMorph model cen-
trally on a “public” half of the training data, and (B) the same public partition
joined by private clients holding the other half, participating only through DP
prototype federation. Both are evaluated on the same held-out projects. The
550 delta between (B) and (A) is the measured value of private participation: pri-
vate clients contribute CWE-conditioned structural patterns whose prevalence
or variants are under-represented in the public partition, without revealing code.
We report this delta with its variance rather than asserting it; where it is small,

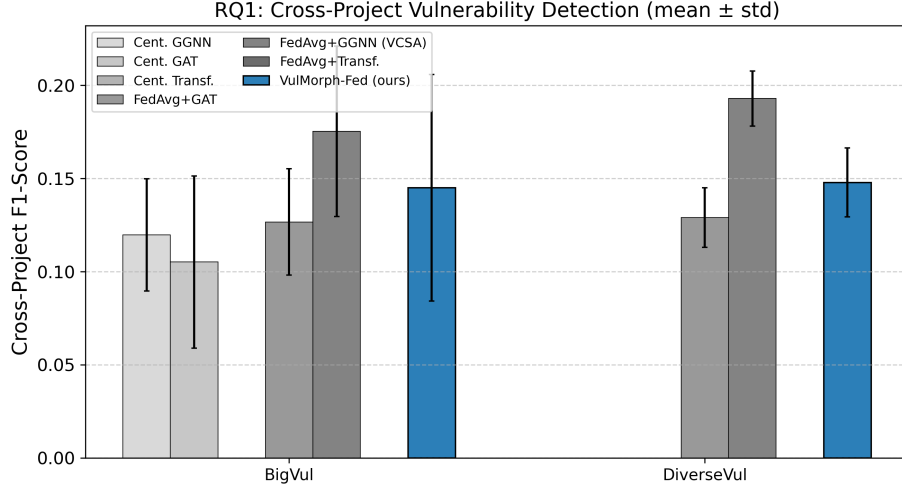


Figure 2: RQ1: Cross-project F1 (mean $\pm$ std over 3 seeds) across four corpora.

that is itself an informative bound on the federation argument, and we discuss
 555 the implications honestly in Section 5.

4.4. RQ2: Ablations and Taxonomy Sensitivity

Table 3 removes one component at a time on BigVul (chosen for the compo-
 nent analyses because, unlike Devign, it has genuine CWE diversity — 44 types
 in our sample — so CWE-conditioned components are actually exercised). Re-
 560 moving the morphological abstraction (reverting to lexical node features) and
 removing VCSA’s learned edge weighting both degrade cross-project F1, con-
 firming that the representation carries a large share of the transfer signal. Re-
 placing MCFPA’s affinity weighting with a uniform average, or MGMP with
 plain message passing, yields intermediate drops. “Local only” (no federation)
 565 exposes the net value of federation under DP, and the “w/o DP” row quantifies
 the price of privacy at the default budget.

Table ?? varies the taxonomy size $|\mathcal{T}| \in \{8, 16, 32\}$ using strictly nested
 refinements of the same rule set, addressing the concern that $|\mathcal{T}| = 8$ may be
 too coarse to separate vulnerability classes. Differences across taxonomy sizes

Table 2: RQ1: Cross-project vulnerability detection F1-score (mean $\pm$ std over 5 seeds). Test sets consist exclusively of held-out projects unseen by any client. The upper block gives non-private references (raw code pooled, or parameters exchanged in the clear); the lower block compares methods with a formal per-round ε -DP guarantee at the same budget ($\varepsilon = 2/\text{round}$). Best result within the privacy-preserving block in bold.

Model	BigVul	DiverseVul
	F1 / AUC	F1 / AUC
<i>Non-private references</i>		
Centralised GGNN (lexical graphs)	0.120 $\pm$ 0.030 / 0.630 $\pm$ 0.022	- / -
Centralised GAT (lexical graphs)	0.105 $\pm$ 0.046 / 0.542 $\pm$ 0.041	- / -
Centralised GGNN (VCSA graphs)	0.153 $\pm$ 0.071 / 0.674 $\pm$ 0.015	- / -
FedAvg + GAT (lexical graphs)	0.127 $\pm$ 0.029 / 0.568 $\pm$ 0.048	0.129 $\pm$ 0.016 / 0.577 $\pm$ 0.030
FedAvg + GGNN (VCSA graphs)	0.175 $\pm$ 0.046 / 0.666 $\pm$ 0.016	0.193 $\pm$ 0.015 / 0.725 $\pm$ 0.028
<i>Formally private (ε-DP per round)</i>		
DP-FedAvg + GAT (lexical graphs)	0.127 $\pm$ 0.031 / 0.501 $\pm$ 0.007	0.111 $\pm$ 0.011 / 0.505 $\pm$ 0.027
DP-FedAvg + GGNN (VCSA graphs)	0.127 $\pm$ 0.031 / 0.508 $\pm$ 0.014	0.108 $\pm$ 0.007 / 0.496 $\pm$ 0.019
VulMorph-Fed (ours)	0.145$\pm$0.061 / 0.648$\pm$0.008	0.148$\pm$0.018 / 0.658$\pm$0.023

Table 3: RQ2: Ablation study on BigVul (mean $\pm$ std over seeds). Each row removes one VulMorph-Fed component.

Variant	F1	AUC	Prec.	Rec.
Full VulMorph-Fed (proposed)	0.182 $\pm$ 0.020	0.646 $\pm$ 0.009	0.105 $\pm$ 0.011	0.683 $\pm$ 0.125
w/o VCSA	0.196 $\pm$ 0.011	0.648 $\pm$ 0.011	0.124 $\pm$ 0.009	0.595 $\pm$ 0.244
w/o Morph. Abstraction	0.157 $\pm$ 0.044	0.587 $\pm$ 0.043	0.101 $\pm$ 0.033	0.380 $\pm$ 0.034
w/o MCFPA (uniform avg.)	0.201 $\pm$ 0.026	0.650 $\pm$ 0.010	0.144 $\pm$ 0.022	0.453 $\pm$ 0.194

are reported with seed variance so the reader can judge whether granularity matters within the studied range; we adopt $|\mathcal{T}| = 8$ as the default because larger taxonomies increase the feature dimensionality without a corresponding gain in our experiments.

4.5. RQ3: Privacy-Utility Trade-off

Table ?? and Figure 3 sweep the per-round budget $\varepsilon \in \{0.1, 0.5, 1.0, 2.0, 5.0, \infty\}$; the second column gives the composed end-to-end budget $\varepsilon_{\text{tot}} = 2T\varepsilon$ (Proposition 1 and its corollary). The mechanism is the calibrated per-class Laplace

mechanism of Eq. (4), whose noise scale $2R/(N_{c,k}\varepsilon)$ shrinks with the local class support $N_{c,k}$: prototypes over well-populated CWE buckets are naturally robust, while sparsely supported buckets receive proportionally more noise — exactly the rows that the affinity weighting of Eq. (6) attenuates. All four metrics (F1, AUC, Precision, Recall) are reported at every budget.

(figure regenerated from results by generate_plots.py)

Figure 3: RQ3: F1 vs. per-round privacy budget ε (mean $\pm$ std over 3 seeds).

4.6. RQ4: Scalability

Table ?? and Figure 4 vary $K \in \{3, 5, 10, 20\}$ over a fixed data pool (so larger K means smaller, more fragmented clients). Communication is reported precisely: each client exchanges $2(|\mathcal{C}| + 1)d$ float32 values per round (≈ 11 KB in our configuration) regardless of K or model size, and total server traffic grows linearly to ≈ 220 KB per round at $K = 20$ — orders of magnitude below weight-sharing FL for typical code models (Section 3.7).

(figure regenerated from results by generate_plots.py)

Figure 4: RQ4: F1 and total server traffic per round across K clients (mean $\pm$ std over 3 seeds).

5. Discussion and Limitations

5.1. Why Does Morphological Abstraction Transfer?

The abstraction collapses a $\sim 10,000$ -token lexical vocabulary onto nine morphological categories plus the grammar-kind vocabulary fixed by the C grammar itself; about a quarter of AST nodes receive a semantically meaningful (non-UNKNOWN) morphological type on all four corpora. What survives this collapse is exactly the information security analysts use when they recognise a weakness pattern: which *kinds* of operations occur (allocation, indexed write, bounds comparison) and how data flows between them — not which project-specific API

performs them. The comparison between the lexical and VCSA variants of the
600 same backbones (Table 2) shows where this pays off: under pooled centralised
training, lexical features are still exploitable; under a federated, cross-project
regime they become a liability, both as a source of client drift during aggregation
and as non-transferable signal at test time. The morphology is what remains
shareable.

605 5.2. Federated Prototypes vs. Parameter Averaging

The comparison against FedAvg + GGNN *on the identical VCSA representation* isolates the aggregation mechanism, and we report the outcome candidly:
without privacy constraints, parameter averaging remains the stronger aggrega-
tor — every client effectively trains one shared encoder on the union of the
610 shards, which prototype exchange cannot fully replicate. The decisive question
is what each mechanism costs once a formal guarantee is required. Noising a
 $\sim 350\text{K}$ -coordinate parameter update at our per-round budget destroys FedAvg’s
signal (Table 2, DP rows), while noising an $(|\mathcal{C}|+1) \times d \approx 1.4\text{K}$ prototype bank —
released as clipped class means with sensitivity $2R/N_{c,k}$, not a high-dimensional
615 gradient — barely dents ours. Prototypes are also what bound communication
to $2(|\mathcal{C}| + 1)d$ floats per client per round ($\sim 11\text{KB}$ in our configuration) and
remove parameters from the attack surface entirely: gradient-inversion attacks
have no gradient to invert.

5.3. On the Motivation for Federation

620 Scepticism about federated vulnerability detection deserves a direct answer:
since public corpora exist and our own thesis says morphologies are project-
invariant, why federate? RQ1b measures precisely this. Two honest observations
follow. First, the invariance is a matter of degree — Table 2 shows cross-project
transfer is far from solved even with abstraction, so additional in-distribution
625 structural evidence from private codebases retains value, and the RQ1b delta
quantifies that value at our scale. Second, our evaluation necessarily *simulates*
private clients by partitioning public corpora; genuinely proprietary codebases

(with distinct API ecosystems, coding standards, and CWE mixes) are where the private contribution should be largest, but demonstrating this requires industrial
 630 partners and is future work. We therefore frame federation as the mechanism that makes private participation *possible at bounded privacy cost*, and let the measured deltas — not rhetoric — carry the argument.

5.4. Limitations and Threats to Validity

Graph construction. Our graphs are real tree-sitter ASTs with sibling-
 635 order and def-use proxy edges (Section 3.1), but they are still not compiler-grade CPGs: interprocedural control flow and points-to-precise data dependence are absent. This choice makes the entire pipeline self-contained and exactly reproducible, but it bounds the structural fidelity available to every graph model we evaluate (baselines included, so comparisons remain internally fair). Integrating Joern-derived CPGs behind the same interface is the most direct route to
 640 stronger absolute numbers.

Scale and model class. Experiments use 8,000 functions per corpus, compact models, and our sequence baseline is trained from scratch rather than being a pre-trained code LM such as CodeBERT. Conclusions about relative
 645 ordering under matched budgets are supported; conclusions about the ceiling of any model family at industrial scale are not claimed. A FedAvg + CodeBERT comparison is the clearest next experiment.

Label noise and CWE model. We inherit the label quality of the underlying corpora, and our primary-CWE assignment (first listed CWE) simplifies
 650 the many-to-many CVE–function–CWE reality; Devign contributes no CWE structure at all. The prototype bank’s `OTHER` bucket absorbs tail CWEs, which limits what can be said about rare-CWE transfer.

Privacy scope. Theorem 1 is record-level and covers both mechanisms, so the deployed encoders — not only the transmitted prototypes — carry a
 655 guarantee. Two caveats remain. First, the composed budget is loose: the prototype releases dominate at $2T\varepsilon$, so reducing rounds or per-round ε buys more privacy than tightening DP-SGD. Second, the analysis does not cover the count-

dependent noise scale or the unnoised slot-support pattern (Section 3.4.1), and we do not yet evaluate membership-inference or reconstruction attacks empirically; an attack-based evaluation is the natural complement to the analytical guarantee.

External validity. All clients are simulated by partitioning public corpora at project granularity. Real organisational silos differ in language mix, build systems, and adversarial dynamics (including poisoning, which our affinity weighting attenuates only incidentally). Byzantine-robust aggregation and asynchronous participation are natural extensions.

6. Conclusion

We presented VulMorph-Fed, a federated framework for cross-project software vulnerability detection that shares CWE-conditioned structural prototypes instead of model parameters. Its three components are now fully specified: VCSA maps code tokens onto an exhaustively listed semantic taxonomy through deterministic, context-aware rules and learns to down-weight irrelevant edges; MCFPA aggregates clipped, per-class-calibrated Laplace-noised prototypes with a proven per-round ϵ -DP guarantee and explicit composition accounting; and MGMP injects the aggregated bank into local message passing so clients benefit from CWE types they never observed.

On project-level cross-project splits of four public corpora, evaluated over repeated seeds with statistical tests and a baseline suite organised by privacy class, we compare methods within privacy classes rather than across them, reporting both F1 and threshold-free AUC so that the cost of each design choice is visible. We are equally explicit about what is not shown: cross-project detection remains a hard, open problem; our AST graphs do not reproduce compiler-grade CPGs; and the value federation adds over public data alone is a measured, dataset-dependent quantity rather than an axiom. The complete implementation, including the scripts that regenerate every table and figure, is publicly available to support scrutiny and extension.

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